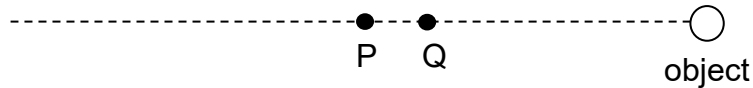


- 10** Two points P and Q are located a fixed distance apart on a straight line joining them to an object considered as a point mass.



The two points P and Q are moved closer to the point mass, while keeping the separation between them constant.

What happens to the magnitudes of their individual gravitational potentials and to the magnitude of the gravitational field strength between the two points?

		magnitudes of gravitational potentials	magnitudes of gravitational field strength difference	
	A	both decrease	decreases	
	B	both decrease	increases	
	C	both increase	stays the same	
	D	both increase	increases	

Answer: D

Gravitational potential at a point is given by:

$$\phi = \frac{-GM}{r}$$

As both P and Q move closer to the mass, their distances r decrease, and since ϕ becomes more negative, the magnitudes of the potentials $|\phi|$ increase.

Gravitational field strength is given by:

$$g = \frac{GM}{r^2}$$

As r decreases, g increases at both P and Q. Because g changes more rapidly with r nearer the mass (steeper gradient), the field strength difference between the two points also increases, even though their separation remains constant.

- 11** A gas molecule of mass m moves with velocity v and collides elastically with a wall perpendicular